ia_study1_suppleAnalyses

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```r
#setwd(dirname(rstudioapi::getActiveDocumentContext()$path))
library(tidyverse)
library(emmeans)
library(psych)
library(patchwork)
source("https://raw.githubusercontent.com/vyqlua/avi_functions/refs/heads/main/ipsatize_fn.R")
```

```r
sensitivity_analyses <- c("!is.na(subID)", "exclude_sd != 1")
```

Load Data

```
data <- read_csv("ia_study1_anon_data_computed.csv")
data <- data[ , -1]

levelBreakdown <- read_csv("ia_TrialsBreakdown.csv")
```

Supple For: Are There Cultural Differences in Ideal Affect between European Americans and Taiwanese?

Raw Ideal Affect Scores

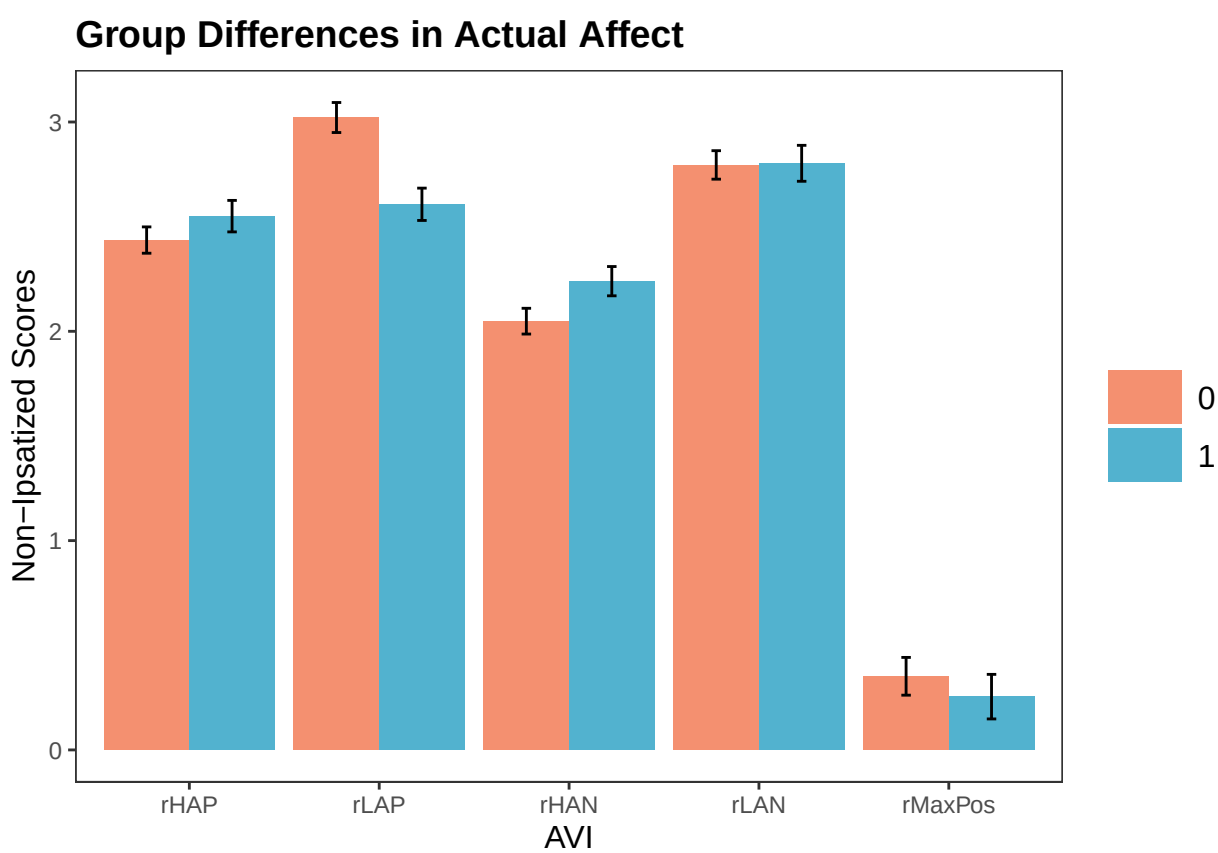
```r
plot_avi(data, group_id = "cultureEA", ipsatized_only = FALSE)
```

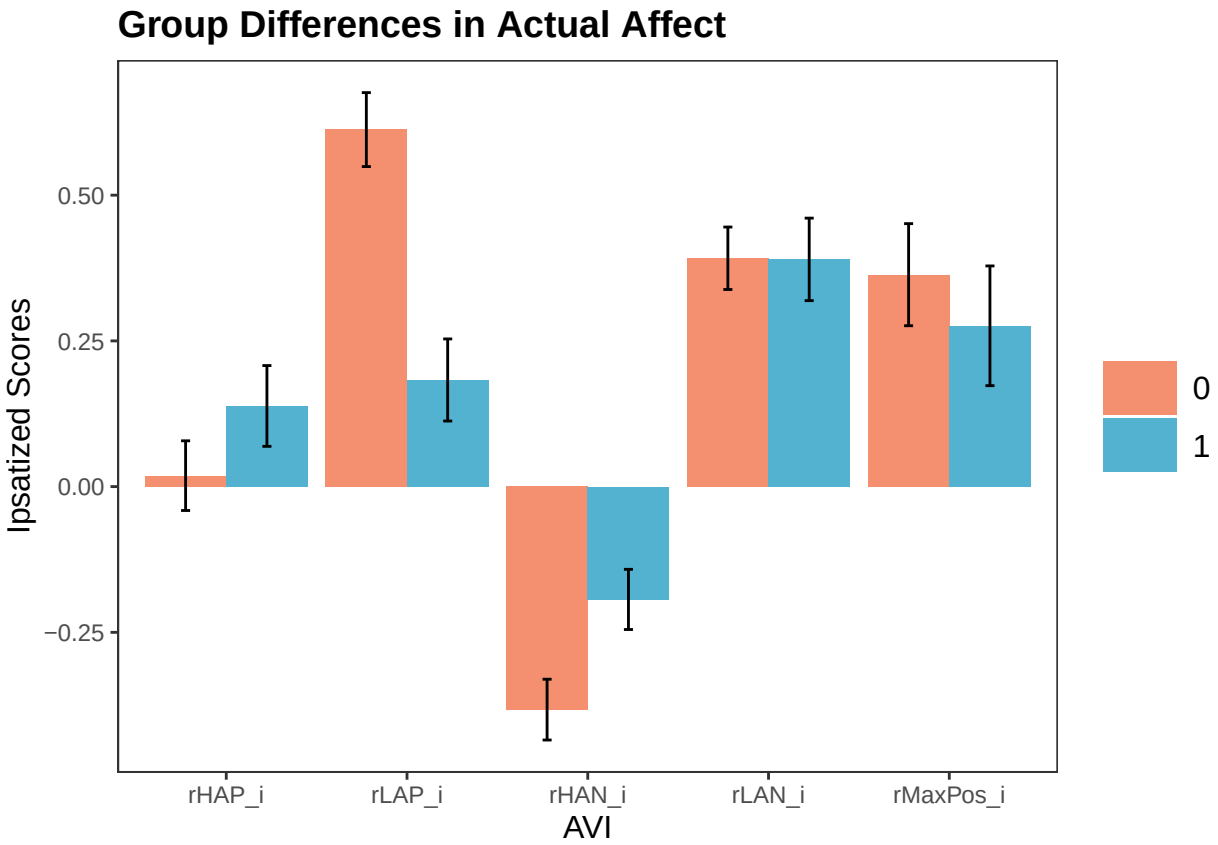
```
## Warning: Using an external vector in selections was deprecated in tidyselect 1.1.0.
## i Please use ‘all_of()‘ or ‘any_of()‘ instead.
##   # Was:
##   data %>% select(group_id)
##
##   # Now:
##   data %>% select(all_of(group_id))
##
## See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

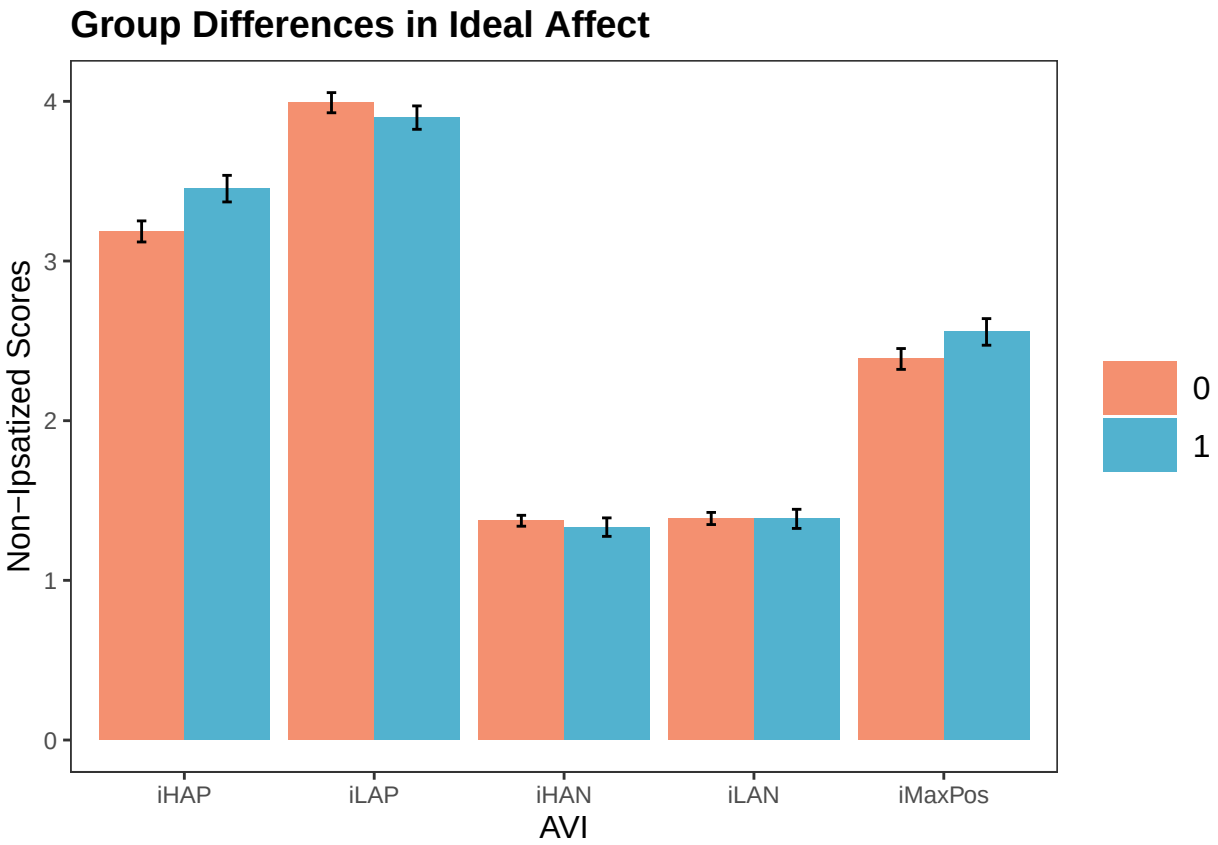
```
## ‘summarise()‘ has grouped output by ’cultureEA’. You can override using the
## ‘.groups‘ argument.
```

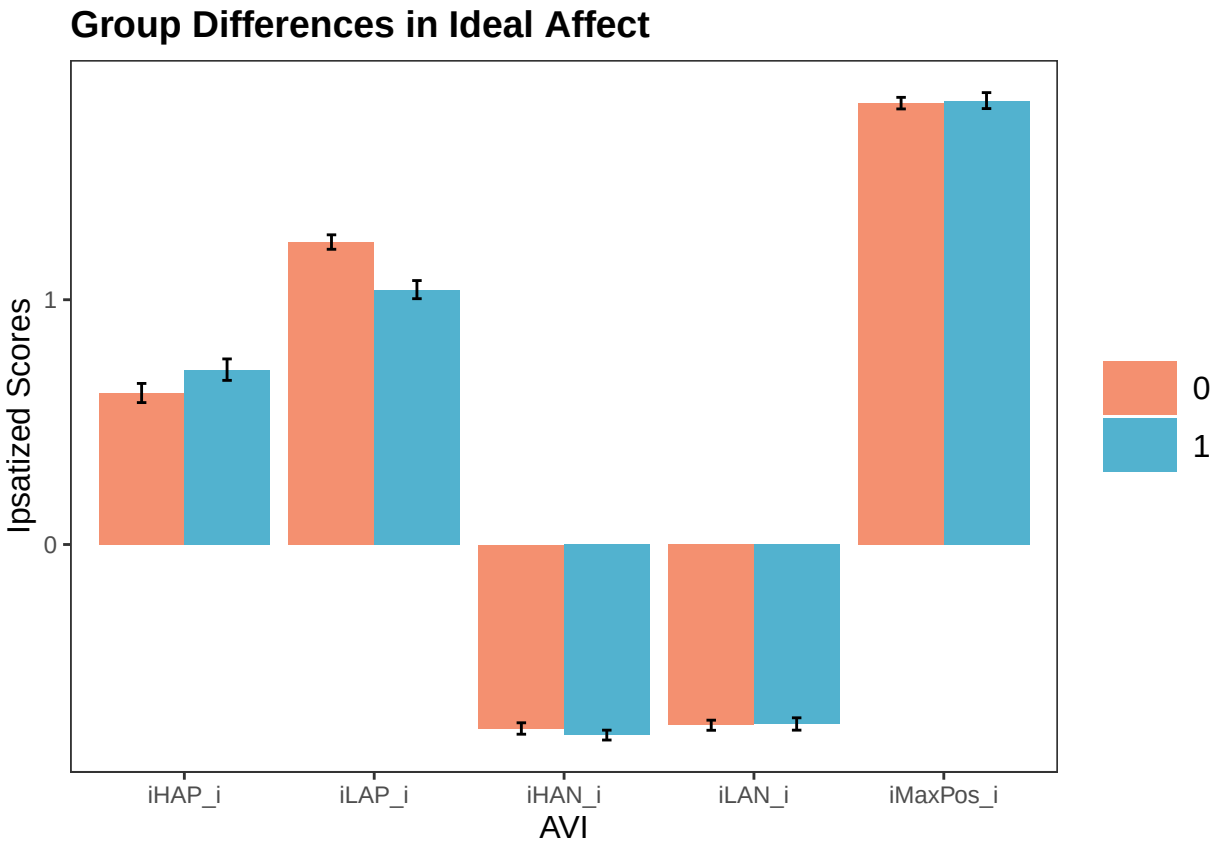
```
## # A tibble: 20 x 6
## # Groups:   cultureEA [2]
##    cultureEA AVI          mean    sd     se AVI_order
##    <fct>     <fct>       <dbl> <dbl>  <dbl>     <dbl>
##  1 0         rHAP       2.44   0.680 0.0629         1
##  2 0         rHAP_i     0.0188 0.647 0.0598         1
##  3 1         rHAP       2.55   0.764 0.0752         1
##  4 1         rHAP_i     0.138  0.703 0.0693         1
##  5 0         rLAP       3.02   0.776 0.0717         3
##  6 0         rLAP_i     0.613  0.686 0.0634         3
##  7 1         rLAP       2.61   0.783 0.0772         3
##  8 1         rLAP_i     0.183  0.715 0.0704         3
##  9 0         rHAN       2.05   0.666 0.0615         4
## 10 0         rHAN_i    -0.383  0.564 0.0522         4
## 11 1         rHAN       2.24   0.710 0.0700         4
## 12 1         rHAN_i    -0.194  0.524 0.0516         4
## 13 0         rLAN       2.79   0.736 0.0681         6
## 14 0         rLAN_i     0.392  0.580 0.0536         6
## 15 1         rLAN       2.80   0.871 0.0858         6
## 16 1         rLAN_i     0.390  0.718 0.0708         6
## 17 0         rMaxPos    0.352  0.975 0.0902         7
## 18 0         rMaxPos_i  0.364  0.947 0.0876         7
## 19 1         rMaxPos    0.254  1.08  0.106          7
## 20 1         rMaxPos_i  0.276  1.04  0.103          7
## # A tibble: 20 x 6
## # Groups:   cultureEA [2]
##    cultureEA AVI         mean    sd     se AVI_order
##    <fct>     <fct>      <dbl> <dbl>  <dbl>     <dbl>
```

```
##  1 0         iHAP       3.19  0.713 0.0659         1
##  2 0         iHAP_i     0.619 0.423 0.0391         1
##  3 1         iHAP       3.45  0.845 0.0833         1
##  4 1         iHAP_i     0.714 0.444 0.0438         1
##  5 0         iLAP       3.99  0.683 0.0631         3
##  6 0         iLAP_i     1.24  0.319 0.0295         3
##  7 1         iLAP       3.90  0.743 0.0732         3
##  8 1         iLAP_i     1.04  0.376 0.0370         3
##  9 0         iHAN       1.37  0.369 0.0341         4
## 10 0         iHAN_i    -0.752 0.252 0.0233         4
## 11 1         iHAN       1.33  0.587 0.0578         4
## 12 1         iHAN_i    -0.779 0.205 0.0202         4
## 13 0         iLAN       1.39  0.410 0.0379         6
## 14 0         iLAN_i    -0.738 0.223 0.0207         6
## 15 1         iLAN       1.39  0.608 0.0599         6
## 16 1         iLAN_i    -0.733 0.256 0.0252         6
## 17 0         iMaxPos    2.39  0.706 0.0653         7
## 18 0         iMaxPos_i  1.80  0.256 0.0237         7
## 19 1         iMaxPos    2.56  0.846 0.0833         7
## 20 1         iMaxPos_i  1.81  0.331 0.0326         7
```




Group Differences in Actual Affect
Ipsatized Scores
0.50
0.25
0.00
−0.25
rHAP_i
rLAP_i
rHAN_i
rLAN_i
rMaxPos_i
AVI
0
1


Group Differences in Ideal Affect
Non−Ipsatized Scores
4
3
2
1
0
iHAP
iLAP
iHAN
iLAN
iMaxPos
AVI
0
1


Group Differences in Ideal Affect
Ipsatized Scores
1
0
iHAP_i
iLAP_i
iHAN_i
iLAN_i
iMaxPos_i
AVI
0
1

Supple For: Does Induced Mood Influence Giving Towards Excited, Calm, and Neutral Responders Differently Across Cultures?

Ipsatized Giving Controlling for Age

Admin before analyses

```
# convert AVI data into long form
longGivingdata_i <- data %>%
  dplyr::select(
    subID, cultureEA, order,
    starts_with("raw") & contains("_i"),
    starts_with("exclude"),
    age # covariates
    ) %>%
  pivot_longer(
    cols = `raw1_i`:`raw36_i`,
    names_to = "trial",
    values_to = "response_i") %>%
  dplyr::mutate(
    trial = as.numeric(substr(trial, start = 4, stop = str_length(trial)-2)))
```

```
# add info on incidental affect (mood), face gender, face race, and face emotion for each trial.
longGivingdata_i = longGivingdata_i %>%
  dplyr::mutate(
    mood = levelBreakdown$mood[trial],
    faceGender = levelBreakdown$faceGender[trial],
    faceRace = levelBreakdown$faceRace[trial],
    faceEmotion = levelBreakdown$faceEmotion[trial])

# make variables factors
longGivingdata_i = longGivingdata_i %>%
  dplyr::mutate(
    culture = ifelse(cultureEA == 1, "EA", "TN") %>% as.factor(),
    mood = factor(mood, levels = c("HAP", "HAN","Neutral")),
    faceRace = factor(faceRace, levels = c("White", "Asian")),
    faceGender = factor(faceGender, levels = c("Male", "Female")),
    faceEmotion = factor(faceEmotion, levels = c("Excited", "Calm", "Neutral")),
    subID = factor(subID, unique(subID)))

longGivingdata_i = longGivingdata_i %>%
  dplyr::filter(!is.na(response_i),
                !is.na(age))
```

Sample size

```r
print("n_total:"); longGivingdata_i %>%
  dplyr::summarise(n = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_total:"
```

```
## [1] 183
```

```r
print("n_tw:"); longGivingdata_i %>%
  dplyr::filter(culture == "TN") %>%
  dplyr::summarise(n_tw = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_tw:"
```

```
## [1] 106
```

```r
print("n_ea:"); longGivingdata_i %>%
  dplyr::filter(culture == "EA") %>%
  dplyr::summarise(n_ea = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_ea:"
```

```
## [1] 77
```

Visualizations

Same as main analyses, hence not plotted

Analyses

```r
aov(
  formula = response_i ~ mood*culture*faceEmotion + mood*age*faceEmotion +
    Error(subID/(mood*faceEmotion)),
  data = longGivingdata_i) %>% summary()
```

```
##
## Error: subID
##            Df  Sum Sq   Mean Sq F value Pr(>F)
## culture     1 1.0e-29 1.190e-29   0.038  0.846
## age         1 4.2e-28 4.160e-28   1.314  0.253
## Residuals 180 5.7e-26 3.166e-28
##
## Error: subID:mood
##               Df Sum Sq Mean Sq F value Pr(>F)
## mood           2   25.2  12.624   4.472 0.0121 *
## mood:culture   2    7.0   3.484   1.234 0.2923
## mood:age       2    2.0   0.996   0.353 0.7029
## Residuals    360 1016.2   2.823
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:faceEmotion
##                      Df Sum Sq Mean Sq F value  Pr(>F)
## faceEmotion           2   15.4   7.720   5.104 0.00652 **
## culture:faceEmotion   2   17.2   8.601   5.686 0.00371 **
## faceEmotion:age       2    3.6   1.812   1.198 0.30304
## Residuals           360  544.5   1.513
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:mood:faceEmotion
##                           Df Sum Sq Mean Sq F value  Pr(>F)
## mood:faceEmotion           4  252.0   62.99  63.569 < 2e-16 ***
## mood:culture:faceEmotion   4   15.3    3.83   3.862 0.00411 **
## mood:faceEmotion:age       4    1.0    0.26   0.263 0.90157
## Residuals                720  713.4    0.99
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: Within
##             Df Sum Sq Mean Sq F value Pr(>F)
## Residuals 4941   3792  0.7675
```

```r
pairwise <- aov(
  formula = response_i ~ mood*culture*faceEmotion + mood*age*faceEmotion +
    Error(subID/(mood*faceEmotion)),
  data = longGivingdata_i)
```

Pairwise Main effect mood

```r
# main effect mood
emm_3way_mood <- emmeans(pairwise, ~ mood)
```

```
## Note: re-fitting model with sum-to-zero contrasts

## Warning in (mth$objs[[1]])(object, trms, xlev, grid, ...): Some predictors are
## correlated with the intercept - results may be very biased

## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_mood) # p values
```

```
##  contrast      estimate     SE  df t.ratio p.value
##  HAP - HAN      0.13991 0.0516 360   2.712  0.0192
##  HAP - Neutral  0.13269 0.0516 360   2.572  0.0283
##  HAN - Neutral -0.00723 0.0516 360  -0.140  0.9892
##
## Results are averaged over the levels of: culture, faceEmotion
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
#confint(pairs(emm_3way_mood)) # confidence intervals
```

Main effect faceEmotion

```r
# main effect faceEmotion
emm_3way_faceEmotion <- emmeans(pairwise, ~ faceEmotion)
```

```
## Note: re-fitting model with sum-to-zero contrasts

## Warning in (mth$objs[[1]])(object, trms, xlev, grid, ...): Some predictors are
## correlated with the intercept - results may be very biased

## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_faceEmotion) # p values
```

```
##  contrast          estimate     SE  df t.ratio p.value
##  Excited - Calm     -0.1122 0.0378 360  -2.972  0.0089
##  Excited - Neutral  -0.0497 0.0378 360  -1.317  0.3869
##  Calm - Neutral      0.0625 0.0378 360   1.655  0.2241
##
## Results are averaged over the levels of: mood, culture
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
#confint(pairs(emm_3way_faceEmotion)) # confidence intervals
```

Culture:faceEmotion

```r
emm_3way_faceEmotionCulture <- emmeans(pairwise, ~ culture | faceEmotion)
```

```
## Note: re-fitting model with sum-to-zero contrasts

## Warning in (mth$objs[[1]])(object, trms, xlev, grid, ...): Some predictors are
## correlated with the intercept - results may be very biased

## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_faceEmotionCulture) # p values
```

```
## faceEmotion = Excited:
##  contrast estimate     SE  df t.ratio p.value
##  EA - TN    0.0919 0.0507 360   1.812  0.0709
##
## faceEmotion = Calm:
##  contrast estimate     SE  df t.ratio p.value
##  EA - TN    0.0142 0.0507 360   0.281  0.7790
##
## faceEmotion = Neutral:
##  contrast estimate     SE  df t.ratio p.value
##  EA - TN   -0.1061 0.0507 360  -2.092  0.0371
##
## Results are averaged over the levels of: mood
```

```r
#confint(pairs(emm_3way_faceEmotionCulture)) # confidence intervals
```

Mood:faceEmotion

```r
emm_3way_faceEmotionMood <- emmeans(pairwise, ~ faceEmotion | mood)
```

```
## Note: re-fitting model with sum-to-zero contrasts

## Warning in (mth$objs[[1]])(object, trms, xlev, grid, ...): Some predictors are
## correlated with the intercept - results may be very biased

## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_faceEmotionMood) # p values
```

```
## mood = HAP:
##  contrast          estimate     SE   df t.ratio p.value
##  Excited - Calm     0.07491 0.0574 1034   1.305  0.3926
##  Excited - Neutral  0.36512 0.0574 1034   6.361  <.0001
##  Calm - Neutral     0.29021 0.0574 1034   5.056  <.0001
##
## mood = HAN:
```

```
##  contrast          estimate     SE   df t.ratio p.value
##  Excited - Calm    -0.40551 0.0574 1034  -7.065  <.0001
##  Excited - Neutral -0.69143 0.0574 1034 -12.046  <.0001
##  Calm - Neutral    -0.28592 0.0574 1034  -4.981  <.0001
##
## mood = Neutral:
##  contrast          estimate     SE   df t.ratio p.value
##  Excited - Calm    -0.00604 0.0574 1034  -0.105  0.9939
##  Excited - Neutral  0.17716 0.0574 1034   3.087  0.0059
##  Calm - Neutral     0.18320 0.0574 1034   3.192  0.0042
##
## Results are averaged over the levels of: culture
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
#confint(pairs(emm_3way_faceEmotionMood)) # confidence intervals
```

Mood:culture:faceEmotion

```r
emm_3way_bymood <- emmeans(pairwise, ~ culture * faceEmotion | mood)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```
## Warning in (mth$objs[[1]])(object, trms, xlev, grid, ...): Some predictors are
## correlated with the intercept - results may be very biased
```

```r
pairs(emm_3way_bymood) # p values
```

```
## mood = HAP:
##  contrast                 estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited  0.005209 0.1037 1176   0.050  1.0000
##  EA Excited - EA Calm     0.050590 0.0957 1034   0.528  0.9950
##  EA Excited - TN Calm     0.104437 0.0979 1066   1.067  0.8944
##  EA Excited - EA Neutral  0.332465 0.0957 1034   3.473  0.0071
##  EA Excited - TN Neutral  0.402986 0.0979 1066   4.117  0.0006
##  TN Excited - EA Calm     0.045381 0.0979 1066   0.464  0.9973
##  TN Excited - TN Calm     0.099228 0.0796 1034   1.246  0.8138
##  TN Excited - EA Neutral  0.327257 0.0979 1066   3.343  0.0111
##  TN Excited - TN Neutral  0.397777 0.0796 1034   4.996  <.0001
##  EA Calm - TN Calm        0.053846 0.1037 1176   0.519  0.9954
##  EA Calm - EA Neutral     0.281875 0.0957 1034   2.944  0.0387
##  EA Calm - TN Neutral     0.352395 0.0979 1066   3.600  0.0045
##  TN Calm - EA Neutral     0.228029 0.0979 1066   2.330  0.1831
##  TN Calm - TN Neutral     0.298549 0.0796 1034   3.750  0.0026
##  EA Neutral - TN Neutral  0.070520 0.1037 1176   0.680  0.9841
##
## mood = HAN:
##  contrast                 estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited  0.114430 0.1037 1176   1.104  0.8798
##  EA Excited - EA Calm    -0.315085 0.0957 1034  -3.291  0.0132
##  EA Excited - TN Calm    -0.381507 0.0979 1066  -3.898  0.0014
##  EA Excited - EA Neutral -0.439463 0.0957 1034  -4.591  0.0001
##  EA Excited - TN Neutral -0.828971 0.0979 1066  -8.469  <.0001
##  TN Excited - EA Calm    -0.429515 0.0979 1066  -4.388  0.0002
##  TN Excited - TN Calm    -0.495937 0.0796 1034  -6.229  <.0001
##  TN Excited - EA Neutral -0.553893 0.0979 1066  -5.659  <.0001
##  TN Excited - TN Neutral -0.943401 0.0796 1034 -11.849  <.0001
##  EA Calm - TN Calm       -0.066422 0.1037 1176  -0.641  0.9879
##  EA Calm - EA Neutral    -0.124378 0.0957 1034  -1.299  0.7856
##  EA Calm - TN Neutral    -0.513885 0.0979 1066  -5.250  <.0001
##  TN Calm - EA Neutral    -0.057956 0.0979 1066  -0.592  0.9916
##  TN Calm - TN Neutral    -0.447463 0.0796 1034  -5.620  <.0001
##  EA Neutral - TN Neutral -0.389508 0.1037 1176  -3.758  0.0025
##
## mood = Neutral:
##  contrast                 estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited  0.156035 0.1037 1176   1.505  0.6609
##  EA Excited - EA Calm     0.044327 0.0957 1034   0.463  0.9973
##  EA Excited - TN Calm     0.099629 0.0979 1066   1.018  0.9121
```

```
##  EA Excited - EA Neutral  0.254887 0.0957 1034   2.663  0.0838
##  EA Excited - TN Neutral  0.255474 0.0979 1066   2.610  0.0957
##  TN Excited - EA Calm    -0.111708 0.0979 1066  -1.141  0.8640
##  TN Excited - TN Calm    -0.056405 0.0796 1034  -0.708  0.9810
##  TN Excited - EA Neutral  0.098852 0.0979 1066   1.010  0.9147
##  TN Excited - TN Neutral  0.099440 0.0796 1034   1.249  0.8124
##  EA Calm - TN Calm        0.055302 0.1037 1176   0.534  0.9948
##  EA Calm - EA Neutral     0.210560 0.0957 1034   2.199  0.2388
##  EA Calm - TN Neutral     0.211147 0.0979 1066   2.157  0.2590
##  TN Calm - EA Neutral     0.155258 0.0979 1066   1.586  0.6079
##  TN Calm - TN Neutral     0.155845 0.0796 1034   1.957  0.3678
##  EA Neutral - TN Neutral  0.000587 0.1037 1176   0.006  1.0000
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```r
#confint(pairs(emm_3way_bymood)) # confidence intervals
```

```r
emm_3way_byfaceEmotion <- emmeans(pairwise, ~ culture * mood | faceEmotion)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```
## Warning in (mth$objs[[1]])(object, trms, xlev, grid, ...): Some predictors are
## correlated with the intercept - results may be very biased
```

```r
pairs(emm_3way_byfaceEmotion) # p values
```

```
## faceEmotion = Excited:
##  contrast                 estimate     SE   df t.ratio p.value
##  EA HAP - TN HAP          0.005209 0.1037 1176   0.050  1.0000
##  EA HAP - EA HAN          0.597629 0.1123  837   5.324  <.0001
##  EA HAP - TN HAN          0.712058 0.0956 1195   7.446  <.0001
##  EA HAP - EA Neutral      0.146911 0.1123  837   1.309  0.7803
##  EA HAP - TN Neutral      0.302946 0.0956 1195   3.168  0.0196
##  TN HAP - EA HAN          0.592420 0.0956 1195   6.195  <.0001
##  TN HAP - TN HAN          0.706850 0.0934  837   7.571  <.0001
##  TN HAP - EA Neutral      0.141703 0.0956 1195   1.482  0.6761
##  TN HAP - TN Neutral      0.297737 0.0934  837   3.189  0.0185
##  EA HAN - TN HAN          0.114430 0.1037 1176   1.104  0.8798
##  EA HAN - EA Neutral     -0.450717 0.1123  837  -4.015  0.0009
##  EA HAN - TN Neutral     -0.294683 0.0956 1195  -3.081  0.0256
##  TN HAN - EA Neutral     -0.565147 0.0956 1195  -5.910  <.0001
##  TN HAN - TN Neutral     -0.409112 0.0934  837  -4.382  0.0002
##  EA Neutral - TN Neutral  0.156035 0.1037 1176   1.505  0.6609
##
## faceEmotion = Calm:
##  contrast                 estimate     SE   df t.ratio p.value
##  EA HAP - TN HAP          0.053846 0.1037 1176   0.519  0.9954
##  EA HAP - EA HAN          0.231953 0.1123  837   2.066  0.3061
##  EA HAP - TN HAN          0.165531 0.0956 1195   1.731  0.5114
##  EA HAP - EA Neutral      0.140648 0.1123  837   1.253  0.8103
##  EA HAP - TN Neutral      0.195950 0.0956 1195   2.049  0.3152
##  TN HAP - EA HAN          0.178107 0.0956 1195   1.862  0.4260
```

```
##  TN HAP - TN HAN          0.111684 0.0934  837   1.196  0.8387
##  TN HAP - EA Neutral      0.086802 0.0956 1195   0.908  0.9446
##  TN HAP - TN Neutral      0.142104 0.0934  837   1.522  0.6501
##  EA HAN - TN HAN         -0.066422 0.1037 1176  -0.641  0.9879
##  EA HAN - EA Neutral     -0.091305 0.1123  837  -0.813  0.9651
##  EA HAN - TN Neutral     -0.036002 0.0956 1195  -0.376  0.9990
##  TN HAN - EA Neutral     -0.024883 0.0956 1195  -0.260  0.9998
##  TN HAN - TN Neutral      0.030420 0.0934  837   0.326  0.9995
##  EA Neutral - TN Neutral  0.055302 0.1037 1176   0.534  0.9948
##
## faceEmotion = Neutral:
##  contrast                 estimate     SE   df t.ratio p.value
##  EA HAP - TN HAP          0.070520 0.1037 1176   0.680  0.9841
##  EA HAP - EA HAN         -0.174300 0.1123  837  -1.553  0.6300
##  EA HAP - TN HAN         -0.563808 0.0956 1195  -5.896  <.0001
##  EA HAP - EA Neutral      0.069333 0.1123  837   0.618  0.9898
##  EA HAP - TN Neutral      0.069920 0.0956 1195   0.731  0.9781
##  TN HAP - EA HAN         -0.244820 0.0956 1195  -2.560  0.1081
##  TN HAP - TN HAN         -0.634328 0.0934  837  -6.794  <.0001
##  TN HAP - EA Neutral     -0.001187 0.0956 1195  -0.012  1.0000
##  TN HAP - TN Neutral     -0.000600 0.0934  837  -0.006  1.0000
##  EA HAN - TN HAN         -0.389508 0.1037 1176  -3.758  0.0025
##  EA HAN - EA Neutral      0.243633 0.1123  837   2.170  0.2528
##  EA HAN - TN Neutral      0.244220 0.0956 1195   2.554  0.1098
##  TN HAN - EA Neutral      0.633141 0.0956 1195   6.621  <.0001
##  TN HAN - TN Neutral      0.633728 0.0934  837   6.788  <.0001
##  EA Neutral - TN Neutral  0.000587 0.1037 1176   0.006  1.0000
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```r
#confint(pairs(emm_3way_byfaceEmotion)) # confidence intervals
```

Raw Giving Scores

Admin before analyses

```r
# Define the analysis function

aov_analysis <-
  function(formula, data, filter_condition1, filter_condition2 = NULL)
    {
    # Initialize a list to store results from each iteration
    analysis_result <- list()
  if (is.null(filter_condition2)) {
    for (i in 1:length(filter_condition1)) {
      subset_data <- data %>%
        filter(eval(parse(text = filter_condition1[i])))
    # Run the analysis on the subset
    analysis_result[[filter_condition1[i]]] <-
      aov(formula, data = subset_data) %>% summary()
    }}
  else {
```

```r
    for (j in 1:length(filter_condition2)) {
      for (i in 1:length(filter_condition1)) {
        subset_data <- data %>%
          filter(eval(parse(text = filter_condition1[i]))) %>%
          filter(eval(parse(text = filter_condition2[j])))
    # Run the analysis on the subset
    analysis_result[[
      paste(filter_condition1[i], "&", filter_condition2[j])]] <-
      aov(formula = formula, data = subset_data) %>% summary()
      }}
    }
  return(analysis_result)
}
```

```r
# convert AVI data into long form
longGivingdata <- data %>%
  dplyr::select(
    subID, cultureEA, order,
    starts_with("raw") & !contains("_i") & !contains("Page"),
    starts_with("exclude"),
    age # covariates
    ) %>%
  pivot_longer(
    cols = `raw1`:`raw36`,
    names_to = "trial",
    values_to = "response") %>%
  dplyr::mutate(
    trial = as.numeric(substr(trial, start = 4, stop = str_length(trial))))
```

```r
# add info on incidental affect (mood), face gender, face race, and face emotion for each trial.
longGivingdata = longGivingdata %>%
  dplyr::mutate(
    mood = levelBreakdown$mood[trial],
    faceGender = levelBreakdown$faceGender[trial],
    faceRace = levelBreakdown$faceRace[trial],
    faceEmotion = levelBreakdown$faceEmotion[trial])

# make variables factors
longGivingdata = longGivingdata %>%
  dplyr::mutate(
    culture = ifelse(cultureEA == 1, "EA", "TN") %>% as.factor(),
    mood = factor(mood, levels = c("HAP", "HAN","Neutral")),
    faceRace = factor(faceRace, levels = c("White", "Asian")),
    faceGender = factor(faceGender, levels = c("Male", "Female")),
    faceEmotion = factor(faceEmotion, levels = c("Excited", "Calm", "Neutral")),
    subID = factor(subID, unique(subID)))

longGivingdata = longGivingdata %>% dplyr::filter(!is.na(response))
```

Sample size

1. Supple analyses (!is.na(subID)) - Includes all participants

```r
print("n_total:"); longGivingdata %>%
  dplyr::summarise(n = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_total:"
```

```
## [1] 220
```

```r
print("n_tw:"); longGivingdata %>%
  dplyr::filter(culture == "TN") %>%
  dplyr::summarise(n_tw = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_tw:"
```

```
## [1] 117
```

```r
print("n_ea:"); longGivingdata %>%
  dplyr::filter(culture == "EA") %>%
  dplyr::summarise(n_ea = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_ea:"
```

```
## [1] 103
```

2. Supple analyses (exclude_sd != 1) - Excludes participants who gave equally to all targets

```r
print("n_total:"); longGivingdata %>%
  dplyr::filter(exclude_sd != 1) %>%
  dplyr::summarise(n = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_total:"
```

```
## [1] 184
```

```r
print("n_tw:"); longGivingdata %>%
  dplyr::filter(exclude_sd != 1, culture == "TN") %>%
  dplyr::summarise(n_tw = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_tw:"
```

```
## [1] 107
```

```r
print("n_ea:"); longGivingdata %>%
  dplyr::filter(exclude_sd != 1, culture == "EA") %>%
  dplyr::summarise(n_ea = dplyr::n_distinct(subID)) %>%
  pull()
```

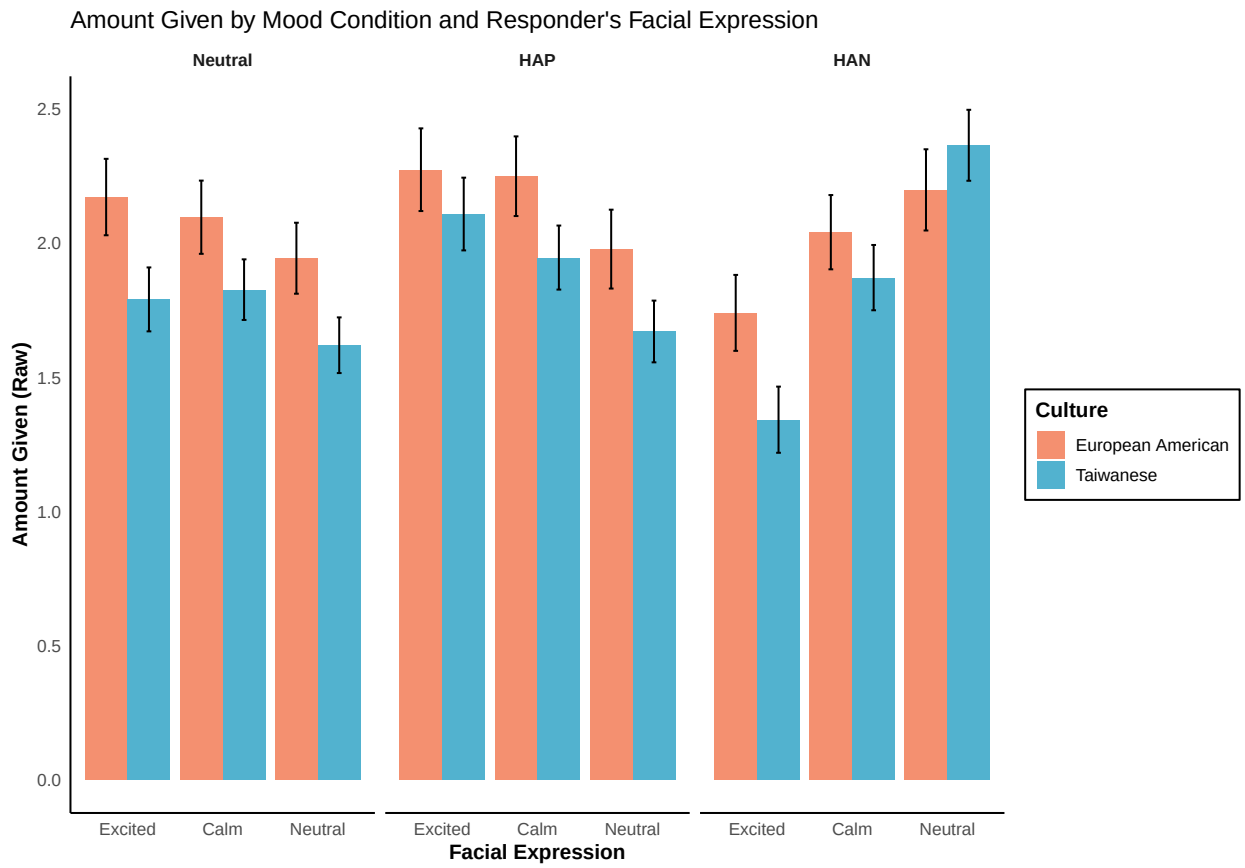
```
## [1] "n_ea:"
```

```
## [1] 77
```

Visualizations

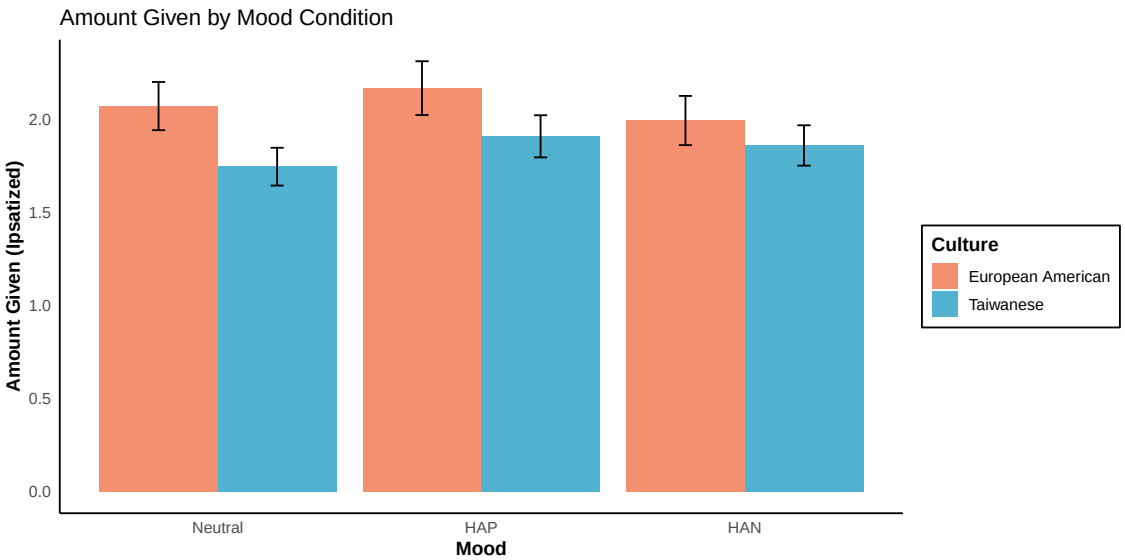
1. Supple analyses (!is.na(subID)) - Includes all participants

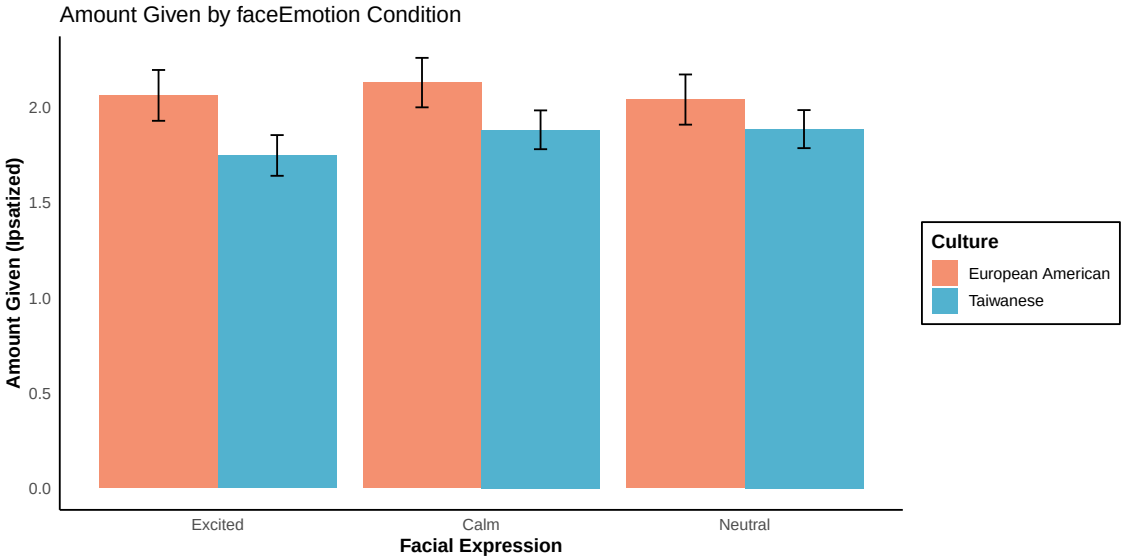
```
longGivingdata2 <- longGivingdata %>% dplyr::mutate(
  culture = if_else(culture == "EA", "European American", "Taiwanese")) %>%
  dplyr::rename(Culture = culture) %>%
  dplyr::mutate(
    mood = factor(mood,
                  labels = c("Neutral", "HAP", "HAN"),
                  levels = c("Neutral", "HAP", "HAN")))
ggplot(
  longGivingdata2 %>%
    group_by(subID, mood, faceEmotion, Culture) %>%
    summarise(response = mean(response, na.rm = T)) %>%
    group_by(faceEmotion, mood, Culture) %>%
    summarise(mean = mean(response, na.rm = T),
              SD = sd(response, na.rm = T),
              SE = sd(response, na.rm = T)/sqrt(n())),
  aes(faceEmotion, mean, fill = Culture)) +
  facet_grid(~mood)+
  scale_fill_manual(
    values = c("European American"="#F49070", "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1)+
  xlab("Facial Expression") +
  ylab("Amount Given (Raw)") +
  ggtitle("Amount Given by Mood Condition and Responder's Facial Expression")
```

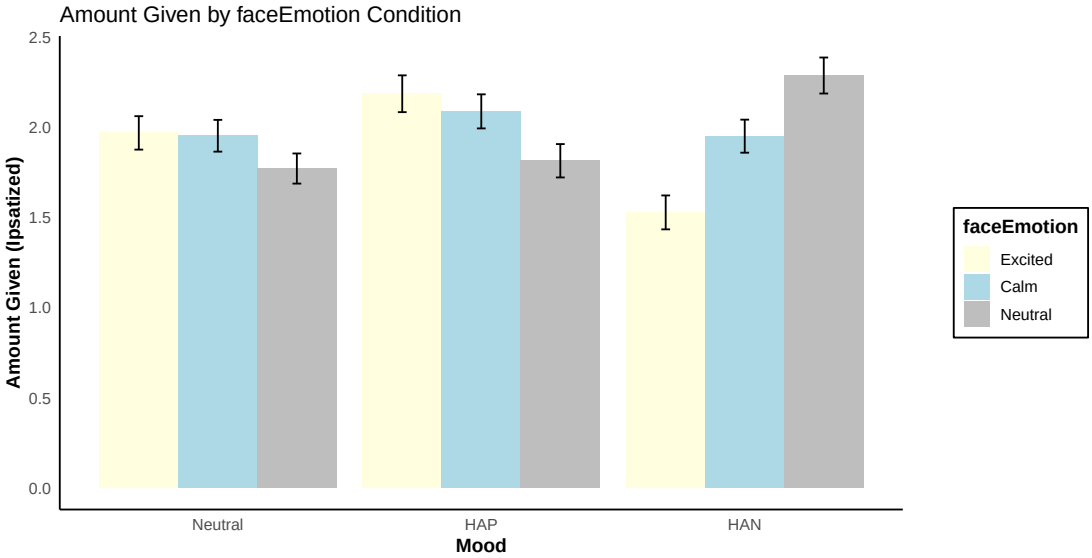



```
p1 <- ggplot(longGivingdata2 %>%
               group_by(subID, mood, Culture) %>%
               summarise(response = mean(response, na.rm = T)) %>%
               group_by(mood, Culture) %>%
               summarise(
                 mean = mean(response, na.rm=T),
                 SD = sd(response, na.rm = T),
                 SE = sd(response, na.rm = T)/sqrt(n())),
             aes(mood, mean, fill = Culture)) +
  scale_fill_manual(values =
                      c("European American"="#F49070",
                        "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1) +
  xlab("Mood") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by Mood Condition")

p2 <- ggplot(longGivingdata2 %>%
               group_by(subID, faceEmotion, Culture) %>%
               summarise(response = mean(response, na.rm = T)) %>%
               group_by(faceEmotion, Culture) %>%
               summarise(
                 mean = mean(response, na.rm=T),
```

```
                 SD = sd(response, na.rm = T),
                 SE = sd(response, na.rm = T)/sqrt(n())),
             aes(faceEmotion, mean, fill = Culture)) +
  scale_fill_manual(values =
                      c("European American"="#F49070",
                        "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1) +
  xlab("Facial Expression") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by faceEmotion Condition")

p3 <- ggplot(longGivingdata2 %>%
               group_by(subID, faceEmotion, mood) %>%
               summarise(response = mean(response, na.rm = T)) %>%
               group_by(faceEmotion, mood) %>%
               summarise(
                 mean = mean(response, na.rm=T),
                 SD = sd(response, na.rm = T),
                 SE = sd(response, na.rm = T)/sqrt(n())),
             aes(mood, mean, fill = faceEmotion)) +
  scale_fill_manual(values =
                      c("Neutral"="grey",
                        "Excited"="lightyellow",
                        "Calm"="lightblue")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1) +
  xlab("Mood") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by faceEmotion Condition")

p1 + p2 + p3 +
  patchwork::plot_layout(
    ncol = 1
  )
```

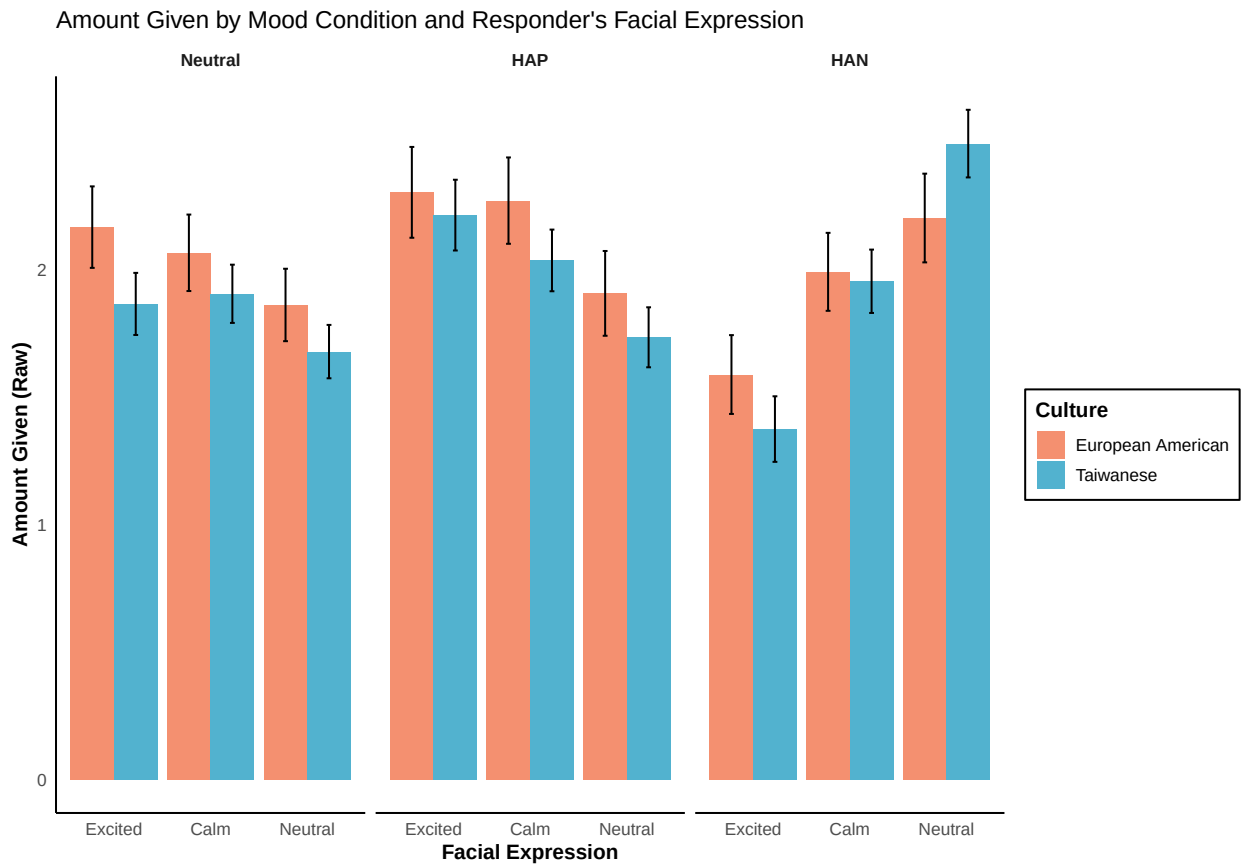

Amount Given by Mood Condition
Amount Given (Ipsatized)
2.0
1.5
1.0
0.5
0.0
Neutral
HAP
HAN
Mood
Culture
European American
Taiwanese


Amount Given by faceEmotion Condition
Amount Given (Ipsatized)
2.0
1.5
1.0
0.5
0.0
Excited
Calm
Neutral
Facial Expression
Culture
European American
Taiwanese


Amount Given by faceEmotion Condition
Amount Given (Ipsatized)
2.5
2.0
1.5
1.0
0.5
0.0
Neutral
HAP
HAN
Mood
faceEmotion
Excited
Calm
Neutral

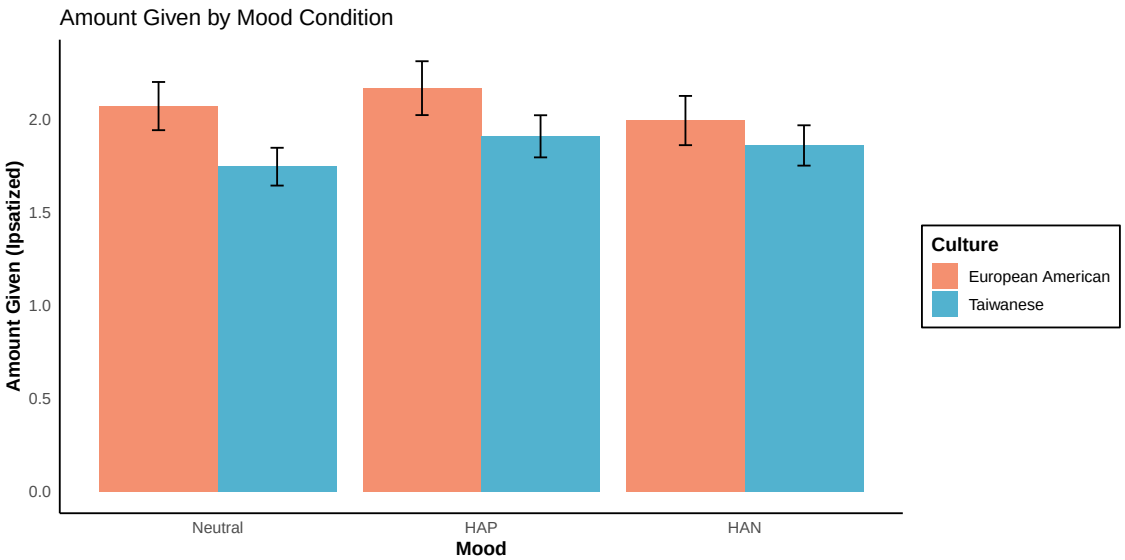
2. Supple analyses (exclude_sd != 1) - Excludes participants who gave equally to all targets

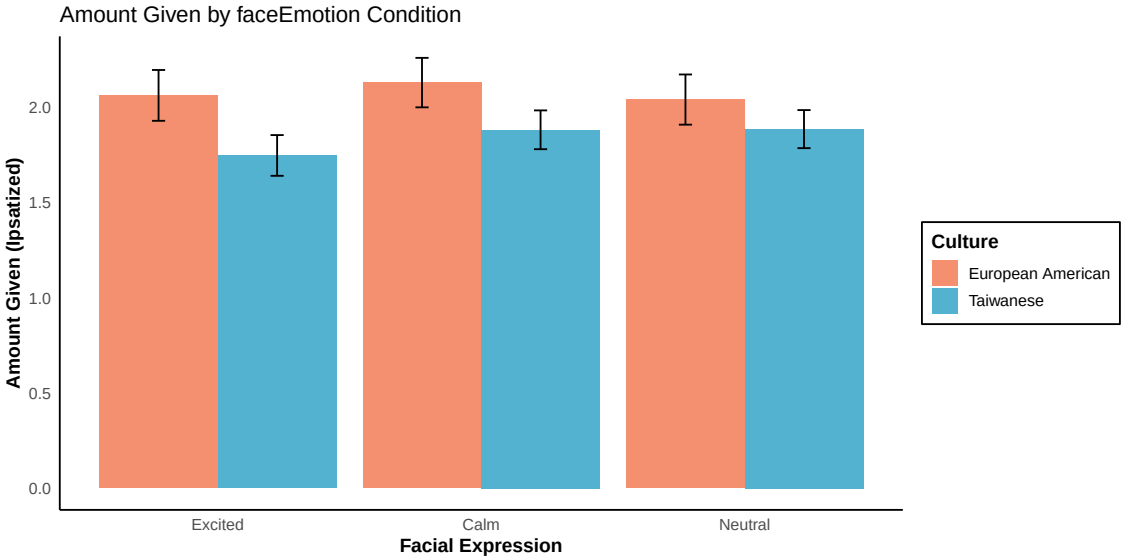
```
longGivingdata2a <- longGivingdata %>%
  dplyr::filter(exclude_sd != 1) %>%
  dplyr::mutate(
  culture = if_else(culture == "EA", "European American", "Taiwanese")) %>%
  dplyr::rename(Culture = culture) %>%
  dplyr::mutate(
    mood = factor(mood,
                  labels = c("Neutral", "HAP", "HAN"),
                  levels = c("Neutral", "HAP", "HAN")))
ggplot(
  longGivingdata2a %>%
    group_by(subID, mood, faceEmotion, Culture) %>%
    summarise(response = mean(response, na.rm = T)) %>%
    group_by(faceEmotion, mood, Culture) %>%
    summarise(mean = mean(response, na.rm = T),
              SD = sd(response, na.rm = T),
              SE = sd(response, na.rm = T)/sqrt(n())),
  aes(faceEmotion, mean, fill = Culture)) +
  facet_grid(~mood)+
  scale_fill_manual(
    values = c("European American"="#F49070", "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1)+
  xlab("Facial Expression") +
  ylab("Amount Given (Raw)") +
  ggtitle("Amount Given by Mood Condition and Responder's Facial Expression")
```

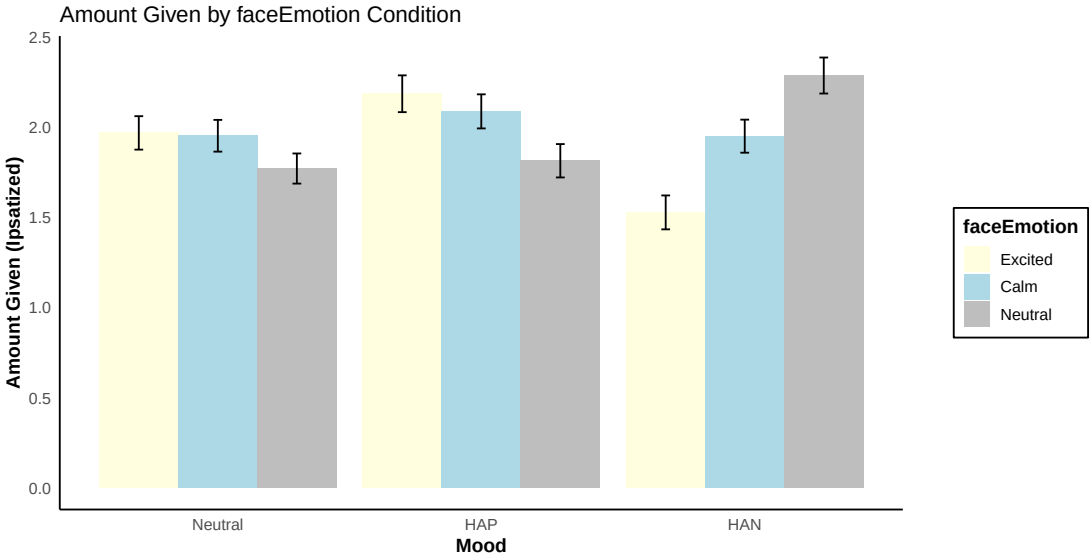



```
p1 <- ggplot(longGivingdata2 %>%
               group_by(subID, mood, Culture) %>%
               summarise(response = mean(response, na.rm = T)) %>%
               group_by(mood, Culture) %>%
               summarise(
                 mean = mean(response, na.rm=T),
                 SD = sd(response, na.rm = T),
                 SE = sd(response, na.rm = T)/sqrt(n())),
             aes(mood, mean, fill = Culture)) +
  scale_fill_manual(values =
                      c("European American"="#F49070",
                        "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1) +
  xlab("Mood") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by Mood Condition")

p2 <- ggplot(longGivingdata2 %>%
               group_by(subID, faceEmotion, Culture) %>%
               summarise(response = mean(response, na.rm = T)) %>%
               group_by(faceEmotion, Culture) %>%
               summarise(
                 mean = mean(response, na.rm=T),
```

```
                 SD = sd(response, na.rm = T),
                 SE = sd(response, na.rm = T)/sqrt(n())),
             aes(faceEmotion, mean, fill = Culture)) +
  scale_fill_manual(values =
                      c("European American"="#F49070",
                        "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1) +
  xlab("Facial Expression") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by faceEmotion Condition")

p3 <- ggplot(longGivingdata2 %>%
               group_by(subID, faceEmotion, mood) %>%
               summarise(response = mean(response, na.rm = T)) %>%
               group_by(faceEmotion, mood) %>%
               summarise(
                 mean = mean(response, na.rm=T),
                 SD = sd(response, na.rm = T),
                 SE = sd(response, na.rm = T)/sqrt(n())),
             aes(mood, mean, fill = faceEmotion)) +
  scale_fill_manual(values =
                      c("Neutral"="grey",
                        "Excited"="lightyellow",
                        "Calm"="lightblue")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1) +
  xlab("Mood") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by faceEmotion Condition")

p1 + p2 + p3 +
  patchwork::plot_layout(
    ncol = 1
  )
```


Amount Given by Mood Condition
Amount Given (Ipsatized)
0.0
0.5
1.0
1.5
2.0
Neutral
HAP
HAN
Mood
Culture
European American
Taiwanese


Amount Given by faceEmotion Condition
Amount Given (Ipsatized)
0.0
0.5
1.0
1.5
2.0
Excited
Calm
Neutral
Facial Expression
Culture
European American
Taiwanese


Amount Given by faceEmotion Condition
Amount Given (Ipsatized)
0.0
0.5
1.0
1.5
2.0
2.5
Neutral
HAP
HAN
Mood
faceEmotion
Excited
Calm
Neutral

Analyses

```r
aov_analysis(
  formula = response ~ mood*culture*faceEmotion +
    Error(subID/(mood*faceEmotion)),
  data = longGivingdata,
  filter_condition1 = sensitivity_analyses)
```

```
## $`!is.na(subID)`
##
## Error: subID
##            Df Sum Sq Mean Sq F value Pr(>F)
## culture     1    113  112.94   2.312   0.13
## Residuals 218  10648   48.84
##
## Error: subID:mood
##               Df Sum Sq Mean Sq F value Pr(>F)
## mood           2   26.0  12.996   2.865  0.058 .
## mood:culture   2   12.4   6.219   1.371  0.255
## Residuals    436 1977.6   4.536
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:faceEmotion
##                      Df Sum Sq Mean Sq F value Pr(>F)
## faceEmotion           2   14.3   7.127   2.889 0.0567 .
## culture:faceEmotion   2    8.5   4.243   1.720 0.1803
## Residuals           436 1075.6   2.467
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:mood:faceEmotion
##                           Df Sum Sq Mean Sq F value   Pr(>F)
## mood:faceEmotion           4  326.9   81.73  57.240  < 2e-16 ***
## mood:culture:faceEmotion   4   31.1    7.78   5.446 0.000247 ***
## Residuals                872 1245.1    1.43
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: Within
##             Df Sum Sq Mean Sq F value Pr(>F)
## Residuals 5940   6105   1.028
##
## $`exclude_sd != 1`
##
## Error: subID
##            Df Sum Sq Mean Sq F value Pr(>F)
## culture     1     24   24.07   0.592  0.443
## Residuals 182   7401   40.67
##
## Error: subID:mood
##               Df Sum Sq Mean Sq F value Pr(>F)
## mood           2   31.1  15.539   2.872 0.0579 .
```

```
## mood:culture   2   15.5   7.764   1.435 0.2395
## Residuals    364 1969.5   5.411
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:faceEmotion
##                      Df Sum Sq Mean Sq F value Pr(>F)
## faceEmotion           2   17.0   8.522   2.893 0.0567 .
## culture:faceEmotion   2    9.1   4.560   1.548 0.2141
## Residuals           364 1072.2   2.946
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:mood:faceEmotion
##                           Df Sum Sq Mean Sq F value Pr(>F)
## mood:faceEmotion           4  390.9   97.72  59.590 <2e-16 ***
## mood:culture:faceEmotion   4   18.4    4.60   2.804  0.025 *
## Residuals                728 1193.8    1.64
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: Within
##             Df Sum Sq Mean Sq F value Pr(>F)
## Residuals 4968   6105   1.229
```

```r
pairwise <- aov(
  formula = response ~ mood*culture*faceEmotion +
    Error(subID/(mood*faceEmotion)),
  data = longGivingdata)
```

Pairwise (Raw Giving, !is.na(subID)) Mood:culture:faceEmotion

```r
emm_3way_bymood <- emmeans(pairwise, ~ culture * faceEmotion | mood)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```r
pairs(emm_3way_bymood) # p values
```

```
## mood = HAP:
##  contrast                estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited   0.1653 0.1865  419   0.887  0.9496
##  EA Excited - EA Calm      0.0243 0.0928 1215   0.262  0.9998
##  EA Excited - TN Calm      0.3277 0.1865  419   1.757  0.4946
##  EA Excited - EA Neutral   0.2961 0.0928 1215   3.191  0.0182
##  EA Excited - TN Neutral   0.6033 0.1865  419   3.236  0.0164
##  TN Excited - EA Calm     -0.1410 0.1865  419  -0.756  0.9745
##  TN Excited - TN Calm      0.1624 0.0871 1215   1.865  0.4245
##  TN Excited - EA Neutral   0.1308 0.1865  419   0.702  0.9817
##  TN Excited - TN Neutral   0.4380 0.0871 1215   5.030  <.0001
##  EA Calm - TN Calm         0.3034 0.1865  419   1.627  0.5809
##  EA Calm - EA Neutral      0.2718 0.0928 1215   2.929  0.0404
##  EA Calm - TN Neutral      0.5791 0.1865  419   3.106  0.0247
##  TN Calm - EA Neutral     -0.0316 0.1865  419  -0.169  1.0000
##  TN Calm - TN Neutral      0.2756 0.0871 1215   3.166  0.0197
##  EA Neutral - TN Neutral   0.3072 0.1865  419   1.648  0.5674
##
## mood = HAN:
##  contrast                estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited   0.3984 0.1865  419   2.137  0.2705
##  EA Excited - EA Calm     -0.3010 0.0928 1215  -3.243  0.0154
##  EA Excited - TN Calm     -0.1315 0.1865  419  -0.705  0.9813
##  EA Excited - EA Neutral  -0.4587 0.0928 1215  -4.943  <.0001
##  EA Excited - TN Neutral  -0.6251 0.1865  419  -3.352  0.0112
##  TN Excited - EA Calm     -0.6994 0.1865  419  -3.751  0.0027
##  TN Excited - TN Calm     -0.5299 0.0871 1215  -6.086  <.0001
##  TN Excited - EA Neutral  -0.8571 0.1865  419  -4.597  0.0001
##  TN Excited - TN Neutral  -1.0235 0.0871 1215 -11.754  <.0001
##  EA Calm - TN Calm         0.1695 0.1865  419   0.909  0.9442
##  EA Calm - EA Neutral     -0.1578 0.0928 1215  -1.700  0.5320
##  EA Calm - TN Neutral     -0.3241 0.1865  419  -1.738  0.5072
##  TN Calm - EA Neutral     -0.3272 0.1865  419  -1.755  0.4962
##  TN Calm - TN Neutral     -0.4936 0.0871 1215  -5.668  <.0001
##  EA Neutral - TN Neutral  -0.1664 0.1865  419  -0.892  0.9483
##
## mood = Neutral:
```

```
##  contrast                estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited   0.3817 0.1865  419   2.047  0.3175
##  EA Excited - EA Calm      0.0752 0.0928 1215   0.811  0.9656
##  EA Excited - TN Calm      0.3454 0.1865  419   1.852  0.4332
##  EA Excited - EA Neutral   0.2282 0.0928 1215   2.458  0.1375
##  EA Excited - TN Neutral   0.5527 0.1865  419   2.964  0.0375
##  TN Excited - EA Calm     -0.3065 0.1865  419  -1.644  0.5699
##  TN Excited - TN Calm     -0.0363 0.0871 1215  -0.417  0.9984
##  TN Excited - EA Neutral  -0.1536 0.1865  419  -0.824  0.9631
##  TN Excited - TN Neutral   0.1709 0.0871 1215   1.963  0.3643
##  EA Calm - TN Calm         0.2702 0.1865  419   1.449  0.6969
##  EA Calm - EA Neutral      0.1529 0.0928 1215   1.648  0.5669
##  EA Calm - TN Neutral      0.4774 0.1865  419   2.561  0.1095
##  TN Calm - EA Neutral     -0.1173 0.1865  419  -0.629  0.9888
##  TN Calm - TN Neutral      0.2073 0.0871 1215   2.380  0.1639
##  EA Neutral - TN Neutral   0.3245 0.1865  419   1.740  0.5058
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```
#confint(pairs(emm_3way_bymood)) # confidence intervals
```

```
emm_3way_byfaceEmotion <- emmeans(pairwise, ~ culture * mood | faceEmotion)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```
pairs(emm_3way_byfaceEmotion) # p values
```

```
## faceEmotion = Excited:
##  contrast                estimate    SE  df t.ratio p.value
##  EA HAP - TN HAP           0.1653 0.186 419   0.887  0.9496
##  EA HAP - EA HAN           0.5340 0.109 966   4.883  <.0001
##  EA HAP - TN HAN           0.9324 0.186 419   5.001  <.0001
##  EA HAP - EA Neutral       0.1019 0.109 966   0.932  0.9382
##  EA HAP - TN Neutral       0.4837 0.186 419   2.594  0.1010
##  TN HAP - EA HAN           0.3687 0.186 419   1.977  0.3571
##  TN HAP - TN HAN           0.7671 0.103 966   7.476  <.0001
##  TN HAP - EA Neutral      -0.0634 0.186 419  -0.340  0.9994
##  TN HAP - TN Neutral       0.3184 0.103 966   3.103  0.0241
##  EA HAN - TN HAN           0.3984 0.186 419   2.137  0.2705
##  EA HAN - EA Neutral      -0.4320 0.109 966  -3.950  0.0012
##  EA HAN - TN Neutral      -0.0503 0.186 419  -0.270  0.9998
##  TN HAN - EA Neutral      -0.8304 0.186 419  -4.454  0.0002
##  TN HAN - TN Neutral      -0.4487 0.103 966  -4.373  0.0002
##  EA Neutral - TN Neutral   0.3817 0.186 419   2.047  0.3175
##
## faceEmotion = Calm:
##  contrast                estimate    SE  df t.ratio p.value
##  EA HAP - TN HAP           0.3034 0.186 419   1.627  0.5809
##  EA HAP - EA HAN           0.2087 0.109 966   1.909  0.3974
##  EA HAP - TN HAN           0.3782 0.186 419   2.028  0.3280
##  EA HAP - EA Neutral       0.1529 0.109 966   1.398  0.7282
##  EA HAP - TN Neutral       0.4231 0.186 419   2.269  0.2092
```

```
##  TN HAP - EA HAN          -0.0947 0.186 419  -0.508  0.9959
##  TN HAP - TN HAN           0.0748 0.103 966   0.729  0.9784
##  TN HAP - EA Neutral      -0.1505 0.186 419  -0.807  0.9661
##  TN HAP - TN Neutral       0.1197 0.103 966   1.166  0.8529
##  EA HAN - TN HAN           0.1695 0.186 419   0.909  0.9442
##  EA HAN - EA Neutral      -0.0558 0.109 966  -0.510  0.9958
##  EA HAN - TN Neutral       0.2143 0.186 419   1.150  0.8602
##  TN HAN - EA Neutral      -0.2253 0.186 419  -1.208  0.8328
##  TN HAN - TN Neutral       0.0449 0.103 966   0.437  0.9980
##  EA Neutral - TN Neutral   0.2702 0.186 419   1.449  0.6969
##
## faceEmotion = Neutral:
##  contrast                estimate    SE  df t.ratio p.value
##  EA HAP - TN HAP           0.3072 0.186 419   1.648  0.5674
##  EA HAP - EA HAN          -0.2209 0.109 966  -2.020  0.3318
##  EA HAP - TN HAN          -0.3872 0.186 419  -2.077  0.3015
##  EA HAP - EA Neutral       0.0340 0.109 966   0.311  0.9996
##  EA HAP - TN Neutral       0.3585 0.186 419   1.923  0.3897
##  TN HAP - EA HAN          -0.5281 0.186 419  -2.832  0.0544
##  TN HAP - TN HAN          -0.6944 0.103 966  -6.768  <.0001
##  TN HAP - EA Neutral      -0.2732 0.186 419  -1.465  0.6866
##  TN HAP - TN Neutral       0.0513 0.103 966   0.500  0.9962
##  EA HAN - TN HAN          -0.1664 0.186 419  -0.892  0.9483
##  EA HAN - EA Neutral       0.2549 0.109 966   2.330  0.1829
##  EA HAN - TN Neutral       0.5794 0.186 419   3.107  0.0245
##  TN HAN - EA Neutral       0.4212 0.186 419   2.259  0.2135
##  TN HAN - TN Neutral       0.7457 0.103 966   7.267  <.0001
##  EA Neutral - TN Neutral   0.3245 0.186 419   1.740  0.5058
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```
#confint(pairs(emm_3way_byfaceEmotion)) # confidence intervals
```

```r
pairwise <- aov(
  formula = response ~ mood*culture*faceEmotion +
    Error(subID/(mood*faceEmotion)),
  data = longGivingdata %>%
    dplyr::filter(exclude_sd != 1))
```

Pairwise (Raw Giving, exclude_sd != 1) Mood:culture:faceEmotion

```r
emm_3way_bymood <- emmeans(pairwise, ~ culture * faceEmotion | mood)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```r
pairs(emm_3way_bymood) # p values
```

```
## mood = HAP:
##  contrast                estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited   0.0893 0.1992  428   0.449  0.9977
##  EA Excited - EA Calm      0.0325 0.1161 1004   0.280  0.9998
##  EA Excited - TN Calm      0.2669 0.1992  428   1.340  0.7624
##  EA Excited - EA Neutral   0.3961 0.1161 1004   3.412  0.0088
##  EA Excited - TN Neutral   0.5683 0.1992  428   2.854  0.0513
##  TN Excited - EA Calm     -0.0569 0.1992  428  -0.286  0.9997
##  TN Excited - TN Calm      0.1776 0.0985 1004   1.803  0.4641
##  TN Excited - EA Neutral   0.3068 0.1992  428   1.540  0.6383
##  TN Excited - TN Neutral   0.4790 0.0985 1004   4.864  <.0001
##  EA Calm - TN Calm         0.2344 0.1992  428   1.177  0.8476
##  EA Calm - EA Neutral      0.3636 0.1161 1004   3.133  0.0220
##  EA Calm - TN Neutral      0.5358 0.1992  428   2.690  0.0792
##  TN Calm - EA Neutral      0.1292 0.1992  428   0.649  0.9871
##  TN Calm - TN Neutral      0.3014 0.0985 1004   3.061  0.0274
##  EA Neutral - TN Neutral   0.1722 0.1992  428   0.865  0.9546
##
## mood = HAN:
##  contrast                estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited   0.2138 0.1992  428   1.074  0.8917
##  EA Excited - EA Calm     -0.4026 0.1161 1004  -3.468  0.0072
##  EA Excited - TN Calm     -0.3656 0.1992  428  -1.836  0.4438
##  EA Excited - EA Neutral  -0.6136 0.1161 1004  -5.286  <.0001
##  EA Excited - TN Neutral  -0.9053 0.1992  428  -4.546  0.0001
##  TN Excited - EA Calm     -0.6164 0.1992  428  -3.095  0.0254
##  TN Excited - TN Calm     -0.5794 0.0985 1004  -5.884  <.0001
##  TN Excited - EA Neutral  -0.8275 0.1992  428  -4.155  0.0006
##  TN Excited - TN Neutral  -1.1192 0.0985 1004 -11.365  <.0001
##  EA Calm - TN Calm         0.0370 0.1992  428   0.186  1.0000
##  EA Calm - EA Neutral     -0.2110 0.1161 1004  -1.818  0.4545
##  EA Calm - TN Neutral     -0.5027 0.1992  428  -2.524  0.1194
##  TN Calm - EA Neutral     -0.2480 0.1992  428  -1.245  0.8141
##  TN Calm - TN Neutral     -0.5397 0.0985 1004  -5.481  <.0001
##  EA Neutral - TN Neutral  -0.2917 0.1992  428  -1.465  0.6870
##
```

```
## mood = Neutral:
##  contrast                estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited   0.3011 0.1992  428   1.512  0.6568
##  EA Excited - EA Calm      0.1006 0.1161 1004   0.867  0.9542
##  EA Excited - TN Calm      0.2614 0.1992  428   1.312  0.7781
##  EA Excited - EA Neutral   0.3052 0.1161 1004   2.629  0.0912
##  EA Excited - TN Neutral   0.4880 0.1992  428   2.450  0.1416
##  TN Excited - EA Calm     -0.2004 0.1992  428  -1.006  0.9157
##  TN Excited - TN Calm     -0.0397 0.0985 1004  -0.403  0.9986
##  TN Excited - EA Neutral   0.0041 0.1992  428   0.021  1.0000
##  TN Excited - TN Neutral   0.1869 0.0985 1004   1.898  0.4038
##  EA Calm - TN Calm         0.1607 0.1992  428   0.807  0.9662
##  EA Calm - EA Neutral      0.2045 0.1161 1004   1.762  0.4909
##  EA Calm - TN Neutral      0.3874 0.1992  428   1.945  0.3762
##  TN Calm - EA Neutral      0.0438 0.1992  428   0.220  0.9999
##  TN Calm - TN Neutral      0.2266 0.0985 1004   2.301  0.1943
##  EA Neutral - TN Neutral   0.1828 0.1992  428   0.918  0.9418
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```r
#confint(pairs(emm_3way_bymood)) # confidence intervals
```

```r
emm_3way_byfaceEmotion <- emmeans(pairwise, ~ culture * mood | faceEmotion)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```r
pairs(emm_3way_byfaceEmotion) # p values
```

```
## faceEmotion = Excited:
##  contrast                estimate    SE  df t.ratio p.value
##  EA HAP - TN HAP           0.0893 0.199 428   0.449  0.9977
##  EA HAP - EA HAN           0.7143 0.137 793   5.208  <.0001
##  EA HAP - TN HAN           0.9281 0.199 428   4.660  0.0001
##  EA HAP - EA Neutral       0.1364 0.137 793   0.994  0.9198
##  EA HAP - TN Neutral       0.4375 0.199 428   2.197  0.2415
##  TN HAP - EA HAN           0.6250 0.199 428   3.138  0.0223
##  TN HAP - TN HAN           0.8388 0.116 793   7.209  <.0001
##  TN HAP - EA Neutral       0.0470 0.199 428   0.236  0.9999
##  TN HAP - TN Neutral       0.3481 0.116 793   2.992  0.0339
##  EA HAN - TN HAN           0.2138 0.199 428   1.074  0.8917
##  EA HAN - EA Neutral      -0.5779 0.137 793  -4.214  0.0004
##  EA HAN - TN Neutral      -0.2768 0.199 428  -1.390  0.7331
##  TN HAN - EA Neutral      -0.7918 0.199 428  -3.975  0.0012
##  TN HAN - TN Neutral      -0.4907 0.116 793  -4.217  0.0004
##  EA Neutral - TN Neutral   0.3011 0.199 428   1.512  0.6568
##
## faceEmotion = Calm:
##  contrast                estimate    SE  df t.ratio p.value
##  EA HAP - TN HAP           0.2344 0.199 428   1.177  0.8476
##  EA HAP - EA HAN           0.2792 0.137 793   2.036  0.3229
##  EA HAP - TN HAN           0.3162 0.199 428   1.588  0.6071
##  EA HAP - EA Neutral       0.2045 0.137 793   1.491  0.6699
```

```
##  EA HAP - TN Neutral       0.3653 0.199 428   1.834  0.4449
##  TN HAP - EA HAN           0.0448 0.199 428   0.225  0.9999
##  TN HAP - TN HAN           0.0818 0.116 793   0.703  0.9816
##  TN HAP - EA Neutral      -0.0299 0.199 428  -0.150  1.0000
##  TN HAP - TN Neutral       0.1308 0.116 793   1.125  0.8712
##  EA HAN - TN HAN           0.0370 0.199 428   0.186  1.0000
##  EA HAN - EA Neutral      -0.0747 0.137 793  -0.544  0.9943
##  EA HAN - TN Neutral       0.0861 0.199 428   0.432  0.9981
##  TN HAN - EA Neutral      -0.1117 0.199 428  -0.561  0.9934
##  TN HAN - TN Neutral       0.0491 0.116 793   0.422  0.9983
##  EA Neutral - TN Neutral   0.1607 0.199 428   0.807  0.9662
##
## faceEmotion = Neutral:
##  contrast                estimate    SE  df t.ratio p.value
##  EA HAP - TN HAP           0.1722 0.199 428   0.865  0.9546
##  EA HAP - EA HAN          -0.2955 0.137 793  -2.154  0.2607
##  EA HAP - TN HAN          -0.5871 0.199 428  -2.948  0.0393
##  EA HAP - EA Neutral       0.0455 0.137 793   0.331  0.9995
##  EA HAP - TN Neutral       0.2283 0.199 428   1.146  0.8617
##  TN HAP - EA HAN          -0.4677 0.199 428  -2.348  0.1774
##  TN HAP - TN HAN          -0.7593 0.116 793  -6.527  <.0001
##  TN HAP - EA Neutral      -0.1267 0.199 428  -0.636  0.9882
##  TN HAP - TN Neutral       0.0561 0.116 793   0.482  0.9968
##  EA HAN - TN HAN          -0.2917 0.199 428  -1.465  0.6870
##  EA HAN - EA Neutral       0.3409 0.137 793   2.486  0.1295
##  EA HAN - TN Neutral       0.5237 0.199 428   2.630  0.0924
##  TN HAN - EA Neutral       0.6326 0.199 428   3.176  0.0198
##  TN HAN - TN Neutral       0.8154 0.116 793   7.009  <.0001
##  EA Neutral - TN Neutral   0.1828 0.199 428   0.918  0.9418
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```
#confint(pairs(emm_3way_byfaceEmotion)) # confidence intervals
```